

Write your questions
and thoughts here!

Recall:

A function's inverse is found by swapping the input (x) and output (y) values!

Confusing Notation:	<u>Reciprocal</u> $x^{-1} =$	or <u>Inverse</u> $f^{-1}(x)$ means
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Three ways to say the same thing:

1. $g(x)$ is the inverse of $f(x)$
2. $g(x) = f^{-1}(x)$
3. $f(g(x)) = x$ and $g(f(x)) = x$

Derivative of an Inverse Function:

$$\frac{d}{dx}[f^{-1}(x)] =$$

The table below gives values of the differentiable functions f , g , and f' at selected values of x . Let $g(x) = f^{-1}(x)$.

x	$f(x)$	$f'(x)$
1	3	-3
2	1	-2
3	-5	-5
4	0	-6

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| 1. What is the value of $g'(1)$? | 2. Write an equation for the line tangent to f^{-1} at $x = 1$. |
| 3. Let g be a differentiable function such that $g(12) = 4$, $g(3) = 6$, $g'(12) = -5$, and $g'(3) = -2$. The function h is differentiable and $h(x) = g^{-1}(x)$ for all x . What is the value of $h'(6)$? | 4. If $f(x) = 3x^3 + 1$ and g is the inverse function of f , what is the value of $g'(25)$? |

3.3 Differentiating Inverse Functions

Calculus

Practice

For each problem, let f and g be differentiable functions where $g(x) = f^{-1}(x)$ for all x .

1. $f(3) = -2$, $f(-2) = 4$,
 $f'(3) = 5$, and $f'(-2) = 1$.
 Find $g'(-2)$.

2. $f(1) = 5$, $f(2) = 4$,
 $f'(1) = -2$, and $f'(2) = -4$.
 Find $g'(5)$.

3. $f(6) = -2$, $f(-3) = 7$,
 $f'(6) = -1$, and $f'(-3) = 3$.
 Find $g'(7)$.

4. $f(-1) = 4$, $f(2) = -3$,
 $f'(-1) = -5$, and $f'(2) = 7$.
 Find $g'(-3)$.

The table below gives values of the differentiable function g and its derivative g' at selected values of x .
 Let $h(x) = g^{-1}(x)$.

x	$g(x)$	$g'(x)$
-1	-2	-4
-2	-5	-2
-3	-4	-1
-4	-3	-5
-5	-1	-3

5. Find $h'(-1)$

Find the equation of the tangent line to g^{-1} at $x = -1$.

6. $h'(-3)$

Find the equation of the tangent line to g^{-1} at $x = -3$.

7. $h'(-5)$

Find the equation of the tangent line to g^{-1} at $x = -5$.

f and g are differentiable functions. Use the table to answer the problems below. f and g are NOT inverses!

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	5	-5	4	5
2	1	-6	3	3
3	6	4	1	6
4	2	9	6	1
5	3	1	1	2
6	4	2	2	4

8. $g^{-1}(4)$

9. $f^{-1}(5)$

10. $\frac{d}{dx}[g^{-1}(x)]_{x=3}$

11. $\frac{d}{dx}[f^{-1}(x)]_{x=1}$

12. Find the line tangent to the graph of $f^{-1}(x)$ at $x = 2$.

For each function $g(x)$, its inverse $g^{-1}(x) = f(x)$. Evaluate the given derivative.

13. $g(x) = \cos(x) + 3x^2$
 $g\left(\frac{\pi}{2}\right) = \frac{3\pi^2}{4}$. Find $f'\left(\frac{3\pi^2}{4}\right)$

14. $g(x) = 2x^3 - x^2 - 5x$
 $g(-2) = -10$. Find $f'(-10)$

15. $g(x) = \sqrt{8 - 2x}$. Find $f'(4)$?

16. $g(x) = x^3 - 7$. Find $f'(20)$?

17. $g(x) = \frac{5}{x+3}$. Find $f'\left(\frac{1}{2}\right)$?

3.3 Differentiating Inverse Functions

18. The functions f and g are differentiable for all real numbers and g is strictly increasing. The table below gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

- (a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

- (b) If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

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19. A function h satisfies $h(3) = 5$ and $h'(3) = 7$. Which of the following statements about the inverse of h must be true?

(A) $(h^{-1})'(5) = 3$

(B) $(h^{-1})'(7) = 3$

(C) $(h^{-1})'(5) = 7$

(D) $(h^{-1})'(5) = \frac{1}{7}$

(E) $(h^{-1})'(7) = \frac{1}{5}$